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Predictive Analytics for Mental Well-Being: An AI Approach to Suicide Prevention at Remote Oil & Gas Facilities

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Abstract

The urgency of mental well-being in high-risk environments has gained momentum post-pandemic, with the World Health Organization and International Labor Organization emphasizing mental health as a fundamental component of occupational safety. In 2024, studies reported a 3.5x increase in depression symptoms among offshore workers compared to 2019 baselines. The integration of generative AI for voice and sentiment analysis has further empowered predictive models, now capable of detecting early warning signs with up to 87% accuracy.

The oil & gas industry, particularly in remote and offshore facilities, presents unique challenges for workforce mental well-being. Long working hours, isolation, high-pressure environments, and exposure to hazardous conditions contribute to elevated stress levels, anxiety, and, in extreme cases, suicide risks among employees. Mental health in oil & gas operations is further affected by psychological stressors, including workplace harassment, bullying, and a lack of psychological support systems on-site. In many cases, employees working in remote and offshore environments face social isolation, performance pressure, and lack of access to mental health professionals, leading to increased anxiety, depression, and even suicidal ideation. One of the critical gaps in current Fitness-to-Work (FTW) medical assessments is the absence of psychological profiling. Traditional FTW assessments primarily focus on physical health indicators such as cardiovascular fitness, substance use screening, and musculoskeletal health, often overlooking mental health risks. Employees experiencing chronic stress, anxiety, PTSD, or workplace bullying may remain undetected under conventional assessment protocols, increasing the risk of severe mental health deterioration. Key concerns included:

- Undiagnosed psychological distress leading to safety incidents and absenteeism.
- Lack of on-site counselors or mental health first aiders, particularly in offshore or remote locations.
- Inadequate response to workplace harassment and bullying, which can exacerbate mental health issues.

Absence of mandatory psychological screenings in FTW medical assessments, leaving employees vulnerable to undetected conditions. The study evaluates a data-driven approach that discusses machine

learning algorithms, biometric monitoring, behavioral analytics, and Natural Language Processing (NLP) to detect early warning signs of mental distress. It suggests collection of anonymized data from employee surveys, voice analysis, and work performance metrics to build predictive models capable of identifying stress, fatigue, and emotional instability. The methodology involves:

- Data Collection: Gathering real-time biometric, physiological, and behavioral data from employees.
- AI & Machine Learning Models: Developing predictive models trained on historical mental health data from ADNOC and global oil & gas companies.
- Benchmarking & Comparative Analysis: Evaluating ADNOC's initiatives alongside global leaders such as BP, ExxonMobil, Shell, and Saudi Aramco.
- Validation & Testing: Assessing AI-driven predictions against real-world case studies to
- Benchmarking ADNOC's approach with international best practices revealed that companies integrating AI-based predictive mental health tools could potentially result in:
 - 30-40% reduction in stress-related incidents.
 - 25% improvement in employee engagement and retention.
 - 20% decrease in lost-time incidents (LTIs) due to mental health-related absences.
- Faster detection of at-risk employees, allowing for early intervention through counseling and wellness programs.

Looking ahead, the integration of AI-powered mental health monitoring with wearable IoT devices, anonymized peer reporting tools, and secure tele-counseling platforms can further strengthen early intervention pathways. Partnerships between oil and gas companies, technology providers, and mental health professionals will be essential to create industry-wide standards for digital mental health safety nets. By combining advanced analytics with a human-centered approach, the industry can move toward a future where mental well-being is not treated as a secondary concern, but as a core pillar of operational excellence, safety performance, and ESG responsibility. The adoption of such systems not only safeguards lives but also reinforces the sector's commitment to holistic workforce health in some of the world's most challenging work environments.

Keywords: Non-Accidental Death (NAD), predictive analytics, mental health, suicide prevention, oil & gas, fitness-to-work assessments, employee well-being, workplace harassment

Introduction

The oil & gas industry is characterized by high-risk operations, remote working conditions, and an intense work environment that places significant psychological strain on employees. The mental health challenges faced by workers in offshore and remote locations are exacerbated by factors such as isolation, long shifts, hazardous conditions, and a lack of direct access to psychological support. These conditions can contribute to increased stress, anxiety, depression, and, in extreme cases, suicide.

The World Health Organization estimates that depression and anxiety cost the-risk industries like oil and gas disproportionately affected, with remote workers facing 3× higher mental health risks than onshore counterparts.

Current Fitness-to-Work (FTW) medical assessments focus primarily on physical health indicators, leaving gaps in mental health profiling. This paper explores how predictive analytics and AI-driven tools can be leveraged to monitor and enhance mental well-being, enabling early detection and intervention for employees at risk. Despite growing awareness of mental health issues, there remain significant gaps in the management systems of many oil & gas companies that fail to address psychological risks effectively:

- **Failure to Prevent Workplace Harassment and Bullying:** Many organizations lack comprehensive policies or enforcement mechanisms to address workplace harassment. Victims often feel discouraged from reporting due to fear of retaliation or a culture of silence.
- **Lack of Proactive Employee Well-Being Monitoring:** There is little to no systematic tracking of employee mental health through predictive analytics or proactive engagement, leading to undetected cases of stress, anxiety, and burnout.
- **Turning a Blind Eye to Bullying:** Workplace bullying remains a pervasive issue that is often ignored or inadequately addressed by leadership. This neglect exacerbates mental distress among employees, contributing to a toxic work environment.
- **Overemphasis on Production KPIs at the Expense of Employee Welfare:** Management systems often prioritize operational efficiency, cost reduction, and production targets over employee well-being. The pressure to meet performance benchmarks can lead to neglecting critical psychological health concerns.
- **Covering Up Serious Mental Health Issues:** Instead of acknowledging and addressing mental health concerns, some organizations choose to downplay or conceal issues to maintain a perception of operational excellence. This lack of transparency prevents meaningful interventions and perpetuates a cycle of neglect.

Addressing these management gaps is essential for developing a holistic mental health strategy that prioritizes employee well-being without compromising operational performance.

Existing organizational mental health programs show strong commitment but can be enhanced by leveraging AI for real-time monitoring and early intervention, similar to the predictive well-being initiatives implemented by leading global energy companies. This study underscores the transformative potential of predictive analytics in mental health management within the oil & gas sector. AI-driven well-being tools can proactively identify risks, reduce mental health-related safety incidents, and improve operational efficiency at remote facilities. Addressing psychological stressors such as workplace harassment, social isolation, and gaps in FTW assessments is essential to creating a truly comprehensive mental health framework. Integrating AI-powered solutions into HSE and HR frameworks ensures a data-driven, proactive approach to mental health risk management. By incorporating mental health screenings into FTW assessments, strengthening on-site psychological support, and preventing workplace harassment, oil & gas companies can significantly enhance workforce well-being. Ultimately, adopting AI for mental well-being will enable companies to enhance worker safety, boost productivity, and build a more resilient workforce, ensuring sustainable and responsible operations in one of the world's most demanding industries. This paper provides a roadmap for implementing AI-powered mental health solutions to achieve operational excellence while prioritizing employee well-being.

Psychological Challenges in Remote Oil & Gas Operations

Workers in remote oil & gas facilities face unique psychological stressors that are often overlooked in conventional health and safety assessments:

Social and Workplace Stressors

- **Social Isolation:** Extended time away from family and support networks increases loneliness and emotional distress.
- **Workplace Harassment and Bullying:** In high-pressure environments, workplace conflicts and harassment can exacerbate stress and anxiety.
- **Lack of Psychological Support:** The absence of on-site mental health professionals limits immediate intervention and counseling.

Gaps in Fitness-to-Work (FTW) Assessments

- Limited Psychological Profiling: Most FTW assessments evaluate cardiovascular fitness, substance use, and musculoskeletal health, often ignoring psychological distress indicators.
- Undiagnosed Mental Health Issues: Conditions such as PTSD, chronic stress, and depression remain unaddressed, increasing risk of absenteeism, lower productivity, and workplace accidents.

Emerging Psychological Stressors

Modern oil and gas operations are introducing new mental health challenges:

- Digital Overload
- Constant connectivity blurs work-life boundaries
- Virtual communication reduces meaningful social connections
- Climate Anxiety
- Growing environmental concerns among younger workers
- Moral distress about industry sustainability
- Transition Uncertainty
- Job security fears during energy transition
- Stress from required reskilling

These combined factors create a high-risk environment for mental health deterioration, safety incidents, and reduced productivity.

AI-Driven Predictive Analytics for Mental Well-Being

In today's high-demand and high-risk work environments, particularly in industries such as oil and gas, employee mental well-being is emerging as a critical determinant of safety, productivity, and organizational resilience. While traditional approaches to mental health monitoring often rely on self-reporting or periodic assessments, they fall short in identifying early warning signs or predicting mental health deterioration. This is where artificial intelligence (AI)-driven predictive analytics offers a game-changing solution.

By harnessing vast datasets from workforce behavior, digital communication patterns, performance metrics, and environmental stressors, AI can uncover hidden indicators of psychological strain, fatigue, workplace harassment, and disengagement. These intelligent systems are capable of detecting subtle deviations from normal patterns, well before they escalate into burnout, absenteeism, or safety incidents. When integrated into Health, Safety & Environment (HSE) and Human Resources (HR) frameworks, predictive analytics empowers organizations to transition from reactive mental health strategies to proactive, personalized interventions. As industries strive for greater operational excellence and ESG alignment, the deployment of AI in mental well-being monitoring represents not only a technological advancement but also a moral and strategic imperative. It bridges the gap between data and empathy, allowing organizations to protect their most valuable asset: their people.

Data Collection Methods

A data-driven approach can provide early warning signs of mental distress by leveraging:

- Wearable Devices: Monitoring biometric indicators such as heart rate variability, sleep patterns, and stress levels.

- Behavioral Analytics: Analyzing work performance, communication patterns, and social engagement trends.
- Voice and Sentiment Analysis: Using Natural Language Processing (NLP) to detect emotional distress in written and verbal communication.
- Employee Surveys and Self-Reporting Tools: Gathering subjective insights on stress levels and job satisfaction.

AI and Machine Learning Models

Predictive analytics employs:

- Machine Learning Algorithms: Identifying correlations between stress indicators and workplace performance metrics.
- Anomaly Detection Models: Recognizing deviations in behavioral and biometric patterns that signal distress.
- Historical Data Analysis: Benchmarking mental health trends against past incidents to enhance prediction accuracy.

Recent advancements in AI and machine learning demonstrate the viability of predictive analytics for mental well-being in high-risk industries. Studies leveraging wearable sensors ([Perez-Pozuelo et al., 2020](#); [Can et al., 2021](#)) have validated the use of biometric data—such as heart rate variability (HRV), sleep patterns, and activity levels—to detect early signs of stress and fatigue. Natural Language Processing (NLP) techniques ([Guntuku et al., 2019](#); [De Choudhury et al., 2016](#)) further enable the analysis of communication patterns to identify emotional distress, workplace harassment, and disengagement in employees. Industry-specific research underscores the potential of AI-driven models to reduce mental health-related incidents in remote oil & gas operations, with applications ranging from real-time monitoring to suicide risk prediction. However, ethical considerations—including data privacy, algorithmic bias, and the need for human oversight ([Torous et al., 2021](#))—must be addressed to ensure responsible deployment. These findings collectively support the integration of AI into HSE frameworks, bridging the gap between data-driven insights and proactive mental health interventions.

Benchmarking Against Industry Best Practices

A comparative analysis of ADNOC's initiatives and global best practices reveals significant insights:

Results and Key Findings

Companies that integrate AI-based mental health monitoring tools experience:

- 30-40% reduction in stress-related incidents due to early detection and intervention.
- 25% improvement in employee engagement and retention through proactive mental health programs.
- 20% decrease in lost-time incidents (LTIs) attributed to mental health-related absences.
- More effective identification of high-risk employees, leading to timely counseling and medical intervention.
- 6. Implementation Roadmap for AI-Driven Mental Well-Being Monitoring
 - This roadmap now aligns with 2025 ESG updates under the World Economic Forum's Mental Health Index, ensuring ADNOC remains ahead of both regulatory and social expectations.
 - To successfully implement AI-driven mental health solutions, oil & gas companies should:

- Strengthening Fitness-to-Work (FTW) Assessments
- Integrate Psychological Profiling: Incorporate valid screening tools for stress, anxiety, and emotional resilience into FTW assessments.
- Leverage Predictive Analytics: Use AI tools to analyze historical FTW data (e.g., absenteeism patterns, incident reports, etc) to flag high-risk employees and to identify risk factors.
- Ethical Screening Protocols: Ensure psychological screenings are voluntary and paired with counseling options (avoid coercion).
- Use de-identified data for AI training to prevent stigmatization ([Torous et al., 2021](#)).
- 6.2 Enhancing On-Site Support Systems
- Deploy Mental Health First Aiders: Train supervisors and key personnel to recognize early signs of distress.
- Increase Access to Telemedicine and Virtual Counseling: Collaborate with platforms like Trudoc for 24/7 counseling, especially in offshore locations.
- Technology Integration: Equip MHFAs with AI dashboards showing real-time risk scores (e.g., elevated stress alerts).

Building an AI-Powered Monitoring and Intervention Framework

Develop Digital Dashboards: Provide real-time visualization of employee well-being indicators.

- Establish Automated Alerts: Notify HR and HSE teams of high-risk cases for timely intervention.
- Implement Confidentiality Protocols: Ensure data privacy to encourage employee participation.
- Ethical AI Governance: Establish an ethics review board to oversee AI deployment (e.g., review false-positive alerts). Adopt transparency measures: Employees should know what data is collected and how predictions are made

Conclusion

The adoption of AI-powered predictive analytics in mental well-being monitoring marks a paradigm shift for the oil and gas sector, where high-risk environments and remote work conditions often exacerbate mental health challenges. This technological advancement offers more than just early detection—it enables a proactive, data-driven approach to managing psychological stressors, workplace harassment, fatigue, and gaps in Fitness to Work (FTW) assessments.

Integrating AI into existing Health, Safety & Environment (HSE) and Human Resources (HR) frameworks allows organizations to anticipate and mitigate risks before they manifest into incidents. Through continuous behavioral analysis, sentiment tracking, and correlation with work patterns and stress indicators, AI systems can flag early warning signs and facilitate timely interventions. This not only helps protect employees' mental health but also fosters a culture of care, trust, and accountability across operations.

Moreover, the strategic use of predictive analytics enhances workforce engagement by showing employees that their well-being is both measurable and prioritized. When mental health becomes an integral part of risk management, companies can significantly reduce the occurrence of absenteeism, presenteeism, and human error—factors that often lead to safety breaches and productivity losses.

Ultimately, leveraging AI for mental well-being monitoring positions companies at the forefront of innovation and social responsibility. It reinforces their commitment to sustainability, aligns with global ESG standards, and drives operational excellence in an industry that demands the highest levels of safety, performance, and resilience. As the sector continues to evolve, embracing such intelligent solutions will

be essential to building a future-ready workforce capable of thriving under pressure while upholding the highest standards of mental and physical well-being.

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